

SYNOPSIS REPORT

CloudCostLens – Smart Cloud Cost Optimization Platform

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1. Introduction

In the modern era of software development, cloud computing platforms such as AWS, Azure, and Google Cloud have become essential for deploying scalable and reliable applications. These platforms offer flexible pricing models, allowing users to pay only for what they use.

However, this flexibility introduces a major challenge - cost unpredictability and lack of visibility. Many users, especially students, startups, and small businesses, often deploy services without fully understanding pricing structures. This results in:

- Unexpected billing
- Resource over-provisioning
- Idle infrastructure costs
- Poor financial planning

CloudCostLens is developed as a cost-awareness and optimization platform that simplifies cloud pricing and provides users with meaningful insights into their spending patterns.

The goal is to bridge the gap between technical deployment and financial understanding, making cloud usage more efficient and controlled.

2. Problem Statement

Despite the growth of cloud computing, cost management remains a major issue due to:

2.1 Lack of Transparency

Cloud providers use complex pricing models involving compute, storage, bandwidth, and region-based costs.

2.2 No Pre-Deployment Cost Awareness

Users often deploy services first and analyze costs later, leading to financial inefficiencies.

2.3 Resource Mismanagement

- Over-provisioned instances
- Idle resources running continuously
- Lack of monitoring

2.4 Limited Tools for Beginners

Existing tools are often:

- Complex
- Enterprise-focused
- Not beginner-friendly

3. Objectives

The CloudCostLens project aims to:

Primary Objectives

- Provide accurate cost estimation based on user inputs
- Deliver clear visualization of cost breakdown
- Help users optimize cloud usage

Secondary Objectives

- Integrate monitoring tools for real-time insights
- Introduce DevSecOps practices
- Build a scalable and deployable web platform

4. Scope of the Project

4.1 Current Scope

- Cost estimation engine
- User input-based calculation
- Web interface for interaction
- Deployment of application

4.2 Extended Scope (In Progress Integration)

- Monitoring using tools like Prometheus and Grafana
- Security checks using SonarQube and Semgrep
- Container orchestration using Kubernetes
- Automation using Ansible

5. Proposed System

CloudCostLens is a web-based analytical platform designed to help users understand and control cloud expenses.

System Capabilities

- Accepts user-defined cloud requirements
- Processes cost calculations using pricing logic
- Displays structured output through dashboards
- Suggests optimization techniques

User Interaction Flow

1. User accesses the web platform
2. Inputs cloud configuration details
3. System processes inputs
4. Cost estimation is generated
5. Optimization suggestions are displayed

6. System Architecture

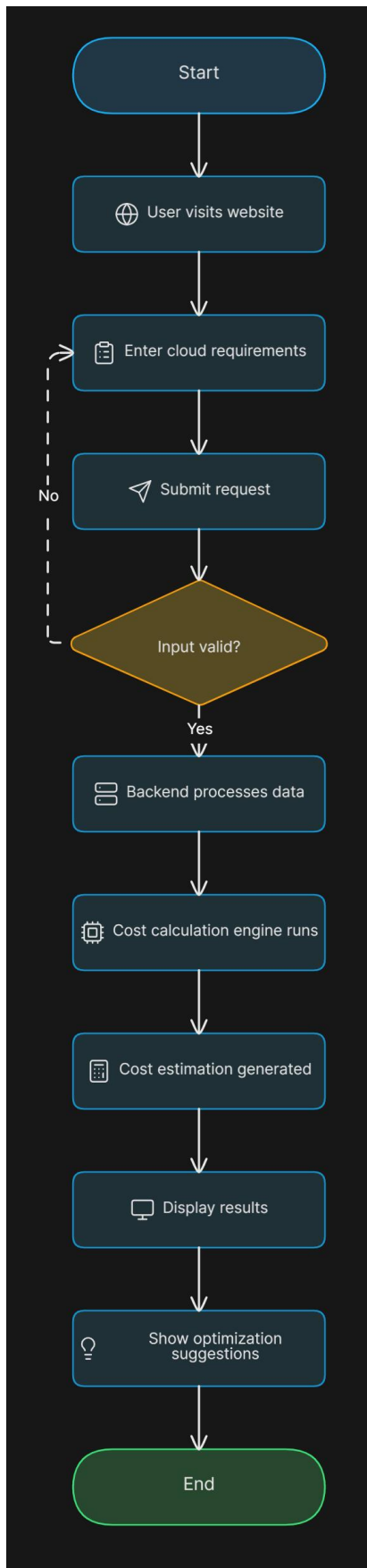
The system follows a multi-layered architecture:

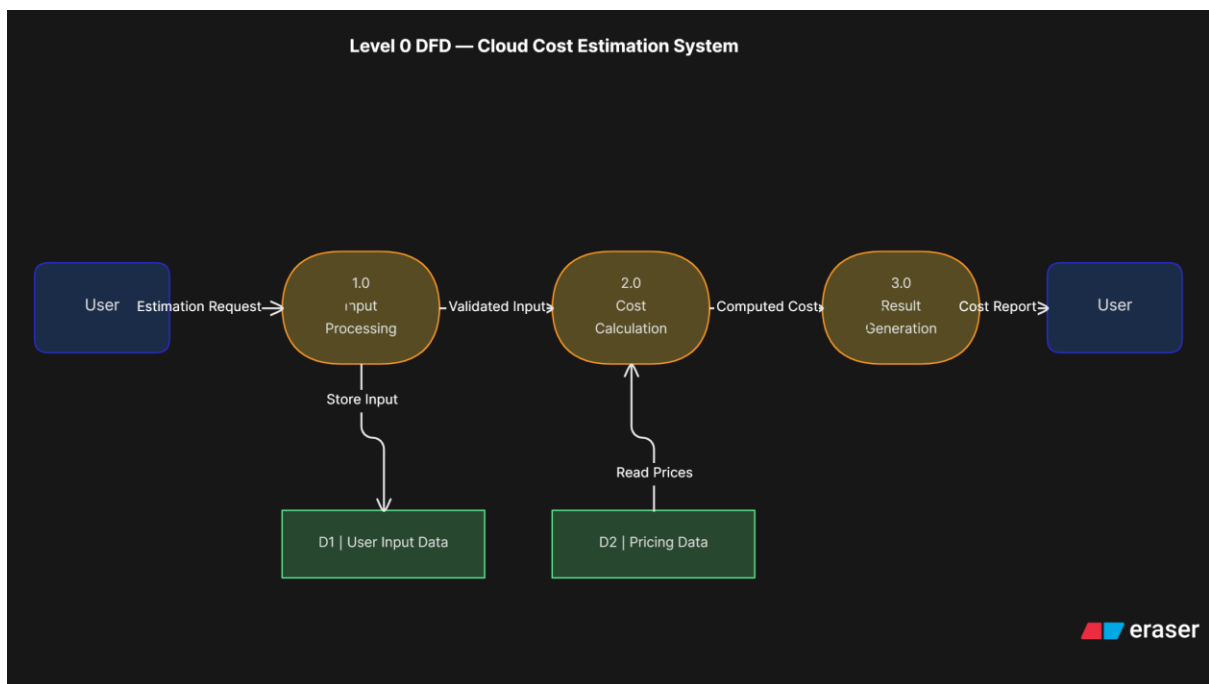
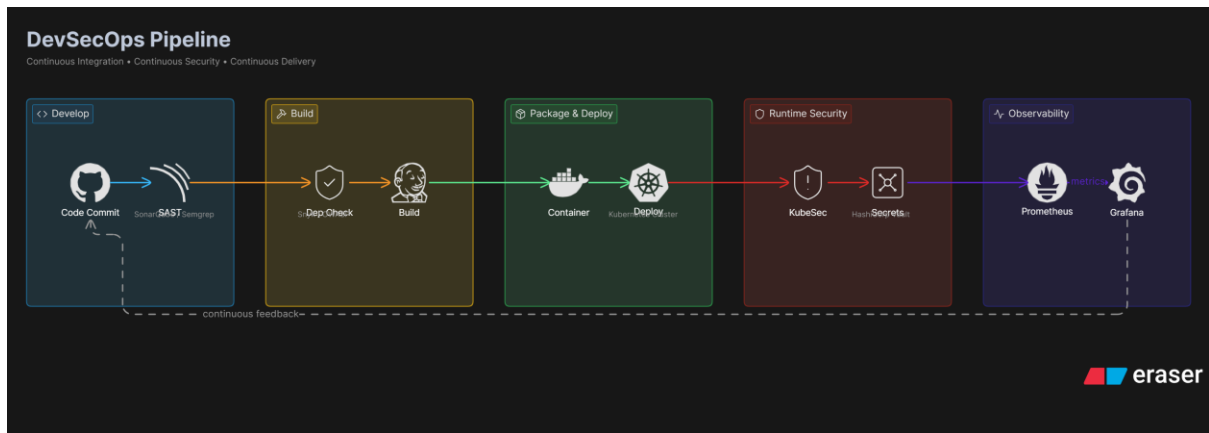
Layers

- **Presentation Layer (Frontend)**
- **Application Layer (Backend APIs)**
- **Logic Layer (Cost Engine)**
- **Data Layer (Storage)**

This architecture ensures:

- Scalability
- Maintainability
- Separation of concerns





7. Features

- User-friendly interface
- Cost breakdown visualization
- Real-time-like estimation
- Optimization recommendations
- Monitoring integration (in progress)
- Security scanning integration

8. Technologies Used

Frontend

- HTML, CSS, JavaScript
- React

Backend

- Java
- SpringBoot

Deployment

- Render

DevOps & Security Tools

- Prometheus
- Grafana
- SonarQube / Semgrep
- OWASP Dependency-Check
- KubeSec
- Kubernetes
- Ansible
- Docker
- Terraform

9. Advantages

- Reduces cloud cost wastage
- Improves financial awareness
- Beginner-friendly platform
- Supports decision-making
- Introduces DevSecOps practices

10. Limitations

- Cost estimation may vary from actual billing
- Limited provider integration
- Monitoring integration under progress

11. Future Enhancements

- AI-based cost prediction
- Multi-cloud comparison
- Real-time billing integration
- User authentication system
- Advanced analytics dashboard

12. Conclusion

CloudCostLens addresses a critical gap in cloud computing by focusing on **cost transparency and optimization**. It simplifies complex pricing models and provides users with actionable insights.

By integrating modern DevOps and security practices, the project not only solves a real-world problem but also aligns with **industry standards**, making it both technically relevant and practically impactful.